

Wet-solvent/oil ratio sensitivity

archived 0.5 vs approved 0.75 and 1.0

Both sensitivity runs reached the time limit without a completed trial. No new product-quality values were obtained. This is not a finding of process infeasibility. The saved S/O = 0.5 design remains unchanged.

Fixed basis: 25 °C; 4000 L/h feed; 1.5 wt% water in wet NMP; fresh-feed NMP and H₂O exactly zero; exactly 10 configured equilibrium stages. Ratio means total wet-solvent mass/oil mass.

Product metrics on explicit mass bases

S/O	Property and basis	Value	Target	Check
0.5	Total aromatics incl. PA (wt% HC)	7.50362	maximum 5	FAIL
0.5	PA subset (wt% HC)	0.15419	maximum 0.5	PASS
0.5	Hydrocarbon recovery (%)	88.70607	minimum 83	PASS
0.5	Saturates (wt% HC)	92.49638	minimum 90	PASS
0.5	NMP (wt% full wet raffinate)	6.38253	maximum 7	PASS
0.75	Timed out; no completed trial or outlet prediction.			
1.0	Timed out; no completed trial or outlet prediction.			

Total aromatics is MONO+DI+POLY+PA on hydrocarbon mass. PA is also shown as its 0.5 wt% subset target. NMP uses the full wet raffinate.

Thermal/numerical gates are independent

S/O	Branch	Raw closure	TPD	Curvature	Phase class	Frozen accepted (raw)	Independent numerical/phase qualification	Overall sulfur
0.75	Timed out; no completed trial or outlet prediction.							
1.0	Timed out; no completed trial or outlet prediction.							

COMPLETED is a file/protocol terminal state, not acceptance. Failed-trial metrics, when present, are retained only as diagnostics. Branch reproduction, raw residual closure, post-split TPD, local curvature, phase distinction, and the frozen worker's raw acceptance are not collapsed into one claim. Product checks use the corrected PA-inclusive aromatic definition; sulfur and physical design acceptance remain unresolved.

Sulfur: three differently based quantities

S/O	Feed-basis proxy (ppm/original HC feed)	Conditional HC-product ppm	Conditional full-wet-product ppm
0.5	1163.2028	1311.3001	1218.1114
0.75	Timed out; no completed trial or outlet prediction.		
1.0	Timed out; no completed trial or outlet prediction.		

The governed allocation is SAT 0, MONO 0.05, DI 0.25, POLY 0.35, PA 0.35 with 3500 ppm feed. The first column is $3500 \times \sum (f_i m_i, R/m_i, F)$, per feed—not product ppm. The two product concentrations each divide that proxy once by their stated product/feed mass ratio. The conversion uses 10^6 , not 10^3 .

Decision boundary and provenance

The 1000 ppm target has no specified product-mass denominator, so neither conditional concentration is silently selected as PASS. This family-allocation post-processing is not species-resolved sulfur transport. No overall validated duty, design, or release claim is made.

The archived 35/40 RPM diameter candidates belong only to S/O 0.5 and are not carried into the 0.75 or 1.0 tests. There is no accepted height, physical compartment count, overall efficiency, final RPM, or final diameter.

Evidence SHA-256

solvent-sensitivity-parent-input-db.json
c0d7c4d863e69adc9f68084b15c63f560369f916c14c6b2d8e403ec9ffef8a25

prepilot-sizing-equilibrium-duty-db.json
cd4309f8a182a32ac3d540cac9fb262d956fde58e20e48711ac99551dd8a61a3

so-0.75:parentCsvWrapperSha256
c0d7c4d863e69adc9f68084b15c63f560369f916c14c6b2d8e403ec9ffef8a25

so-0.75:equilibriumDutyCsvWrapperSha256
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so-0.75:requestSha256
66c36eb0aec10e8389c0d70832c43f11769ffdbca87be3e49ebbb675e825d6be

so-0.75:inputSha256
aab47b3f5e65fa763f64807e51aa4c8946e7ae0d349821b975bc9841b383e619

so-0.75:engineSha256
4f0b57dd41a2d768e1964e49ecee23e87de358ad7de697a2b315ab94a79091ac

so-0.75:modelSha256
f165f0d23eeae810e40341f1cc86a2b97514bc21b74838d4a08eab1760e6e1af

so-0.75:managedManifestSha256
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so-0.75:managedRunnerSha256
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so-0.75:frozenWorkerSourceSha256
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so-0.75:parentStage1SnapshotHash
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so-0.75:runnerStdoutSha256
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so-0.75:runnerStderrSha256
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so-1.0:equilibriumDutyCsvWrapperSha256
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bcf6bf55e6475a01fbf6452d70cfcf4616f61755203ac815ddd381fb2aeef876

so-1.0:inputSha256
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so-1.0:managedRunnerSha256
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so-1.0:frozenWorkerSourceSha256
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so-1.0:parentStage1SnapshotHash
a717f993f983648c72bbbe82c1edcf7a3b3d0f8ec334301aaff24a20a426d3f3

so-1.0:runnerStdoutSha256
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855

so-1.0:runnerStderrSha256
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